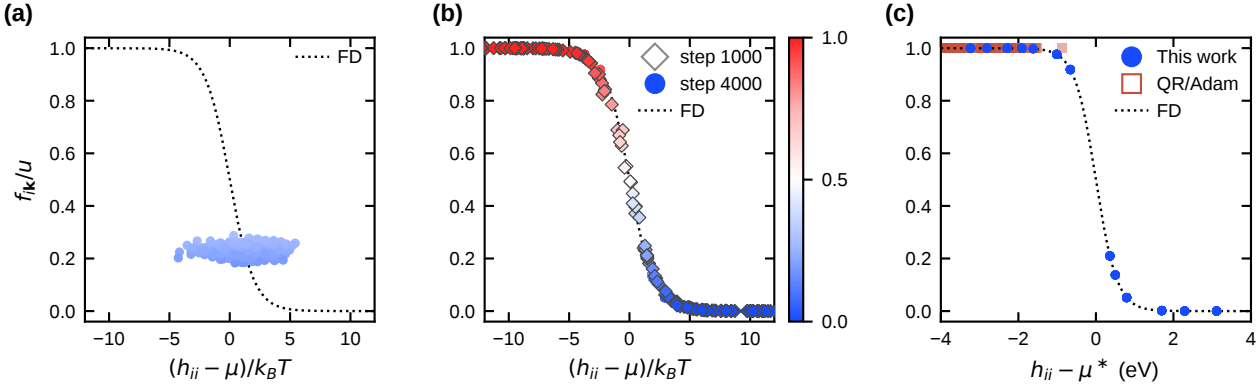

(a)
(b)
(c)
FD
step 1000
step 4000
This work
QR/Adam
$f_{\mathbf{k}}/u$
$(h_{ii}-\mu)/k_BT$
$h_{ii}-\mu^*$ (eV)